Restriction kernels in extensions of string markings

Human Compression project

Abstract

We characterize when adjoining a transposition at a new point preserves a string C-marking. The tests are inclusions between pointwise stabilizers in successive original prefix groups. They do not assume the intersection property of the enlarged marking. Applying the criterion on the two sides of a split gives a necessary and sufficient condition for the rank and degree extension when one internal split factor is trivial. A parity example explains the obstruction. The last end-intersection in the printed proof of Cameron–Fernandes–Leemans can be established directly; the missing condition is the intersection property of its recursively obtained ends. No conclusion about the source's main stability theorem is inferred.

1 Conventions and the source of the obstruction

All permutation sets are finite. The notation $\mathrm{Sym}(X)$ denotes permutations supported on X, with every point outside X fixed. For a group preserving X, its restriction to the complement denotes the actual induced image, extended by the identity on X. A string C-marking consists of distinct nonidentity involutions, with distant marks commuting and with

$$\langle s_j : j \in I\rangle \cap \langle s_j : j \in J\rangle = \langle s_j : j \in I \cap J\rangle$$

for all label subsets. Every subfamily of a C-marking has this property. We retain the standard end criterion: a string marking is C if both end deletions are C and their groups intersect in the middle group. This is McMullen–Schulte, Proposition 2E16, recalled as Proposition 2.1 of [CFL].

The exact source [CFL] is arXiv:2212.12723v2, 11 January 2023. Its Proposition 2.4 asserts an invalid unrestricted converse for removing an endpoint central twist. Its Theorem 6.4 also fails as stated, even for a transitive input. The separate complete counterexamples are not repeated here. The proposed repair concerns the additional group information needed by the construction, not a change in the meaning of its source hypotheses. The pointwise stabilizers below detect correlations which disappear when an induced permutation action is mistaken for an independently supported subgroup.

2 Adjoining one transposition

Lemma 1. *Let $P \le \mathrm{Sym}(\Omega)$, let $a \in \Omega$, and let c be a new point. Extend P to fix c, put $X = a^P$ and $t = (a\ c)$. Then*

$$\langle P, t\rangle = \mathrm{Sym}(X \cup \{c\}) \times P^{\Omega\setminus X}.$$

Proof. Conjugating t by P gives every $(x\ c)$ for $x \in X$. These transpositions generate the full symmetric group on $X \cup \{c\}$: $(x\ c)(y\ c)(x\ c) = (x\ y)$. Cancel the restriction to X of any element of P using that supported symmetric group. What remains is its actual restriction to $\Omega \setminus X$, fixing the other points. Thus both displayed factors are contained in the generated group. Conversely the original generators preserve their supports and have the indicated complementary restrictions. The supports are disjoint, so the product is direct. □

Lemma 2. *Let $P \le Q \le \mathrm{Sym}(\Omega)$ and $X = a^P$. Then*

$$Q \cap \langle P, (a\ c)\rangle = P \quad \Longleftrightarrow \quad Q \cap \mathrm{Sym}(X) \le P.$$

The group $Q \cap \mathrm{Sym}(X)$ is the pointwise stabilizer in Q of $\Omega \setminus X$.

Proof. An element of Q fixes c. By Lemma 1, its membership in the second group on the left means that it preserves X and that its complementary restriction belongs to $P^{\Omega\setminus X}$. Choose $p \in P$ with that same restriction. Multiplying by p^{-1} leaves an element of Q supported on X. The right-hand inclusion therefore implies that the original element belongs to P. The reverse containment is automatic. Conversely every element of $Q \cap \mathrm{Sym}(X)$ belongs to $Q \cap \langle P, (a\ c)\rangle$, so the equality implies the inclusion. □

Theorem 3. *Let $(h_0, \dots, h_{k-1})$ be a string C-marking on Ω, and suppose h_j fixes a for every $j > 0$. Put*

$$P_j = \langle h_0, \dots, h_{j-1}\rangle, \qquad X_j = a^{P_j} \quad (0 \le j \le k),$$

where $P_0 = 1$. For a new point c, the marking

$$((a\ c), h_0, \dots, h_{k-1})$$

is C if and only if

$$P_j \cap \mathrm{Sym}(X_{j-1}) \le P_{j-1} \qquad (1 \le j \le k). \tag{1}$$

The first condition is automatic. For $k = 0$ the enlarged marking has rank one.

Proof. The new transposition commutes with h_j for $j > 0$, since each of those involutions fixes both its points. It is distinct from every old mark because it moves the new point.

Assume (1). Induct on j to show that $((a\ c), h_0, \dots, h_{j-1})$ is C. The initial rank-one case is immediate. At the next stage one end is the old prefix with group P_j, which is C by hypothesis. The other is the preceding extended prefix, which is C by induction. Their intersection is P_{j-1} by Lemma 2 and the jth condition. The end criterion proves the assertion.

Conversely, if the entire new marking is C, every indicated prefix is C. Its end groups must intersect in P_{j-1}. Lemma 2 gives (1) for each j. Since $X_0 = \{a\}$ and $\mathrm{Sym}(\{a\}) = 1$, the first condition is indeed automatic. □

For $k \ge 1$, these are at most $k - 1$ nontrivial tests in groups generated by the original marks. Each is a pointwise-stabilizer computation followed by containment; none asks for an intersection property in the enlarged degree. Equivalently, the two pointwise stabilizers

$$P_{j(\Omega\setminus X_{j-1})} \quad \text{and} \quad P_{j-1(\Omega\setminus X_{j-1})}$$

must be equal. For finite groups equality can be checked by their orders, because the second is a subgroup of the first.

Corollary 4. *Under the hypotheses of Theorem 3, if $h_0 = (a\ b)$ is a pure transposition, adjoining $(a\ c)$ preserves the C-property.*

Proof. For each $j \ge 1$, the subgroup generated by $h_1, \dots, h_{j-1}$ fixes a. Apply Lemma 1 with a as its new point and b as the distinguished old point. It follows that P_j contains $\mathrm{Sym}(X_j)$. Hence every group on the left of (1) is contained in $\mathrm{Sym}(X_{j-1}) \le P_{j-1}$, with the trivial $j = 1$ case treated separately. □

3 Two independent sides and a central twist

For disjoint sets $A, B, \{c\}$, points $a \in A, b \in B$ and groups $L \leq \mathrm{Sym}(A)$, $T \leq \mathrm{Sym}(B)$, write

$$K(L, T) = \langle L, T, (a\ c), (c\ b) \rangle.$$

Lemma 5. *If $L_0 \leq L_1$ and $T_0 \leq T_1$, then*

$$K(L_0, T_1) \cap K(L_1, T_0) = K(L_0, T_0).$$

Proof. Writing $X = a^L$ and $Y = b^T$, conjugation of the two pure transpositions and cancellation of restrictions, as in Lemma 1, give

$$K(L, T) = \mathrm{Sym}(X \cup Y \cup \{c\}) \times L^{A \setminus X} \times T^{B \setminus Y}.$$

Indeed, the star transpositions from both sides generate the first factor; the remaining restrictions are separately available on disjoint sets. Put $X_j = a^{L_j}$ and $Y_j = b^{T_j}$. The two enlarged root sets intersect in $Z = X_0 \cup Y_0 \cup \{c\}$. An element of the intersection preserves Z, acts on $A \setminus X_0$ in the actual restriction image of L_0, and acts on $B \setminus Y_0$ in that of T_0. These three sets partition the whole domain, so the displayed formula puts it in $K(L_0, T_0)$. The reverse inclusion follows from the two containments. □

We also need the valid forward central-extension fact, and include its proof to avoid invoking the invalid converse. This forward statement is Fernandes–Leemans–Mixer (2012), Proposition 5.3, as cited in [CFL, Proposition 2.2] and the gluing source.

Lemma 6. *Let $(s_0, \dots, s_{m-1})$ be a string C-marking of H. In an ambient direct product $H \times \langle z \rangle$ with z of order two, replace just the last mark by $s_{m-1}z$. The resulting marking is C. The same holds for $z = 1$ and for the first endpoint.*

Proof. Projection to H sends each new mark to its old counterpart, preserving distinctness and non-identity. The string commutations are unchanged. Induct on m. The rank-one case is immediate. One end deletion is untwisted and C; the other is an endpoint twist of a shorter C-marking and is C by induction. An element in their intersection projects into the old middle subgroup by the old C-property. Projection is the identity on the untwisted end, so that element itself is its projected middle element. All middle marks are untwisted. The two new ends therefore intersect precisely in their middle group, and the end criterion finishes the induction. Reversal treats the first endpoint. □

4 A criterion for the split extension

Let $(\rho_0, \dots, \rho_{r-1})$ be a string C-marking on $\Omega = A \sqcup B$, with $a \in A, b \in B$, such that

$$\rho_j \text{ is supported on } A\ (j < i), \qquad \rho_j \text{ is supported on } B\ (j > i), \qquad \rho_i = (a\ b)\beta, \tag{2}$$

where β is an involution or the identity supported on $B \setminus \{b\}$. This is the perfect-split form with $\alpha_i = 1$; we do not require transitivity, a fracture graph, or moved endpoints in the following criterion. Introduce a point c and replace ρ_i by

$$u = (a\ c), \qquad v = (c\ b)\beta.$$

Consider the following two original string submarkings, both rooted at a:

$$\mathcal{L} = (\rho_{i-1}, \dots, \rho_0), \qquad \mathcal{R} = (\rho_i, \rho_{i+1}, \dots, \rho_{r-1}).$$

The first may be empty. Apply the prefix notation of Theorem 3 separately to each.

Theorem 7. *The prescribed extended marking is C if and only if both $\mathcal{L}$ and $\mathcal{R}$ satisfy* (1) *at a. There are at most r tests on prefix kernels, at most $r-2$ of them nontrivial when both lists are nonempty. The symmetric construction in which the internal factor on B is trivial is obtained by reversing the order and interchanging the two sides.*

Proof. All marks of $\mathcal{R}$ after its first fix a. For $j < i-1$, commutation of ρ_j with ρ_i forces ρ_j to fix a: evaluate the equality at b, which every left mark fixes. Thus all marks of $\mathcal{L}$ after its first also fix a.

The inserted marks are nonidentity involutions, distinct from each other and from the old marks because they move the new point. The extended marking satisfies the string commutations. To verify the only potentially new ones, commutation of an old right mark with ρ_i at distance at least two forces it to fix b and to commute with β. The old left marks at distance at least two fix a, and they all commute with β by disjoint support. These statements give every distant pair involving u or v.

The kernel tests and Theorem 3 make the left marking $(\rho_0, \dots, \rho_{i-1}, u)$ C, after reversal. They also make $(u, \rho_i, \rho_{i+1}, \dots)$ C. Conjugating the latter by u fixes u and every old right mark, and sends ρ_i to v. Thus the entire right marking $(u, v, \rho_{i+1}, \dots)$ is C.

We verify every consecutive interval of the extended marking by induction on its length. Intervals not containing both u, v are submarkings of the left or right C-markings just identified, or old submarkings, so are C. The same is true of intervals starting at u and extending rightwards. An interval ending at v is C as well: first extend its left part, ending in the pure mark u, by the pure $(c\ b)$, using Corollary 4 after reversal. Then multiply just this last mark by β. Its support is disjoint from the entire preceding construction, so Lemma 6 applies.

It remains to treat an interval having at least one old mark on each side of u, v. Its end deletions are C by induction. Moreover $(uv)^3 = \beta$, since its three-cycle and the disjoint β commute. Hence a selected subgroup containing both u, v is exactly $K(L, T)$, where L is its selected left subgroup and T is generated by β and its selected old right marks. Deleting the first or last mark changes these to $K(L_0, T_1)$ and $K(L_1, T_0)$, with $L_0 \le L_1$, $T_0 \le T_1$. Lemma 5 identifies their intersection with the middle group. The end criterion completes the interval induction and proves the whole marking C.

Conversely, a C extended marking has both displayed left and right submarkings C. Reverse the first and conjugate the second by u. The necessity direction of Theorem 3 then gives exactly the two families of tests on original prefix kernels. □

Corollary 8. *If $\rho_i = (a\ b)$ is itself pure in* (2)*, replacing it by $(a\ c), (c\ b)$ preserves the string C-marking.*

Proof. The old left submarking ending in ρ_i is C; renaming its point b as c gives the left marking needed in the proof above. The right submarking begins with the pure ρ_i and every later mark fixes a, so Corollary 4 gives the right marking. The proof of Theorem 7 therefore applies. □

5 Parity fibers and explicit uses

The criterion can be evaluated from a subgroup description without a full subgroup census. Suppose P preserves X and $Y = \Omega \setminus X$ and has the fiber form

$$P = \{(g, d) \in L \times D : \phi(g) = \psi(d)\},$$

where $L \le \mathrm{Sym}(X)$, $D \le \mathrm{Sym}(Y)$ and ϕ, ψ are homomorphisms to a common group. Then $P \cap \mathrm{Sym}(X) = \ker \phi$, viewed as a supported subgroup. For $P \le Q$, Lemma 2 asks exactly that $Q \cap \mathrm{Sym}(X)$ lie in this kernel. The quotient maps in such a description are the familiar fiber-product data of Goursat's lemma; no classification by that lemma is needed for this direct computation.

In the transitive counterexample the relevant original suffix is

$$q = (12)(34), \quad r = (26)(35)(47), \quad s = (34)(67).$$

It is a C-marking of S_7, as proved in the separate note. For the prefix $P_2 = \langle q, r \rangle$, its root orbit at 1 is $X_2 = \{1, 2, 6\}$. Put $t = (34)$, $w = (35)(47)$ and $D = \langle t, w \rangle \cong D_8$. The written subgroup formula there is

$$P_2 = \{gd : g \in \mathrm{Sym}(X_2),\ d \in D,\ \mathrm{sgn}(g) = \chi(d)\},$$

where $\chi(t) = \chi(w)$ is the nontrivial element of C_2. Consequently $P_2 \cap \mathrm{Sym}(X_2) = \mathrm{Alt}(X_2)$, whereas $P_3 \cap \mathrm{Sym}(X_2) = \mathrm{Sym}(X_2)$ because $P_3 = S_7$. The third prefix test fails with index two. This is the precise lost parity relation, detected in original groups of degrees at most seven.

The test is not a requirement that every original subgroup be an independent direct product. For example, on points $v_0, \dots, v_k$ and a disjoint pair interchanged by z, take

$$h_j = (v_j\ v_{j+1})z \qquad (0 \le j < k).$$

This is the image of the adjacent-transposition marking of S_{k+1} under the injective homomorphism $g \mapsto gz^{\varepsilon(g)}$, so is a string C-marking. Every prefix has this same parity-graph form. At step $j \ge 2$, its kernel on the complement of $X_{j-1} = \{v_0, \dots, v_{j-1}\}$ is precisely $\mathrm{Alt}(X_{j-1})$, equal to the preceding prefix's kernel. Thus a pure new leaf at v_0 preserves C even though the original first mark is not pure. Subsequent pure leaves may be added repeatedly by Corollary 4. These are written uniform deductions, not extrapolations from sample orders.

For the already admitted gluing families, any interior pure edge of the added tail has precisely the separated lower and higher supports of (2). Corollary 8 therefore justifies its subdivision uniformly, with the folded or doubled core left intact. This is a bounded application of the repaired theorem, not a new classification or a proof of all the high-rank cases in [CFL].

6 What is and is not repaired

When both inserted marks occur, their cube already separates β. The product calculation and the end-intersection on p. 33 of [CFL] can therefore be supplied by Lemma 5, without an unproved independence of restriction images. That equality alone was not sufficient: the end criterion also needs both ends to be C, and recursion reaches endpoint attachments at impure marks. Those attachments are governed by Theorem 3; the old C-property and a fracture graph do not force its kernel inclusions. This is why adding transitivity to the printed hypotheses does not repair Theorem 6.4.

The standard end criterion and forward twist theorem are attributed above. The supported-transposition argument reuses the common gluing proof; the new interface isolates the exact additional restriction-kernel condition and its two-sided use. No historical priority assertion is made. Verification of these conditions throughout the main stability range, and recovery of the remaining classification and counting arguments of [CFL], remain separate obligations. The main stability theorem is not claimed false.

The accompanying Lean file verifies three general subgroup statements: the cancellation argument for equal images under a homomorphism, its equivalence with the intersection of a subgroup and an inverse image, and the equivalent equality of kernels. These are actual subgroup statements, valid without finiteness. The permutation restriction maps, the supported symmetric factors and the induction using the end criterion are proved above; they are not formalized by those three declarations. The complete GAP file tests the criterion and its hypotheses in bounded examples. Its finite outputs are diagnostics, not premises of the quantified theorems.

References

[1] P. J. Cameron, M. E. Fernandes and D. Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, 11 January 2023. Exact bound source edition; Propositions 2.1–2.4 and Theorem 6.4 are the locators discussed here.